

False-Positive Triggers in Neural Audio Generation Interfaces: A Root-Cause Analysis and Empirical Testing Framework for Subgenre and Geographic Descriptors

Abstract

Large Language Model (LLM)-driven generative music pipelines employ automated content moderation layers to prevent the generation of illicit, copyrighted, or sensitive material. However, these safety mechanisms frequently produce false positives when processing domain-specific subgenre terminology or geographic metadata. This paper analyzes a case study in which an AI audio generation pipeline repeatedly threw prohibition errors (GRANULAR_FINISH_REASON_PROHIBITED_INPUT_CONTENT) during attempts to generate a 120 BPM "acid jazz" track themed around rain in the Netherlands, before succeeding when restricted to simplified genre parameters. We propose three empirical hypotheses isolating string-matching collisions, geopolitical filtering, and prompt-length overhead, establishing a systematic testing methodology to identify the failure mechanism.

1. Introduction

Text-to-audio foundation models rely on multi-stage pipeline architectures. Before an input prompt reaches the diffusion or autoregressive audio model (e.g., Google's Lyria architecture), it passes through Natural Language Processing (NLP) classifiers and keyword-filtering safety layers. These moderation layers enforce policies regarding controlled substances, copyright infringement, and regional regulatory compliance.

Because automated moderation systems heavily prioritize precision against high-risk categories (such as narcotics or hate speech), they are vulnerable to linguistic collisions with legitimate musical jargon. A primary example is the musical subgenre "acid jazz," which contains the string "acid"—a colloquial reference to lysergic acid diethylamide (LSD). This paper evaluates the structural failure modes of a prompt sequence, investigates the underlying causes of five consecutive execution halts, and provides a structured empirical framework to isolate the trigger.

2. Literature Review

Automated content moderation systems utilized in generative artificial intelligence rely on fine-tuned transformer models and rule-based regular expressions to scan prompts for policy violations. As noted in recent studies on generative AI safety policies, platforms must comply with strict acceptable use rules regarding controlled substances and illegal acts.

However, NLP-based content moderation frequently exhibits false-positive errors when processing domain-specific jargon. In music informatics and streaming platforms, terminology related to subgenres (e.g., *Acid House*, *Acid Jazz*, *Psychedelic Rock*, *Gabba*) regularly collides with drug-related lexicons. Furthermore, automated moderation systems deployed under international regulatory frameworks—such as the European Union’s AI Act—often integrate geographic and regional compliance checks that increase prompt scrutiny when specific country or city identifiers are detected.

3. Case Study and Failure Analysis

3.1 Operational Sequence

- **Invocations 1–5:** Prompts containing the parameters Acid Jazz, 120 BPM, and raining in Netherlands (or Amsterdam) were submitted to the audio generation tool. All five attempts returned a GRANULAR_FINISH_REASON_PROHIBITED_INPUT_CONTENT error.
- **Invocation 6:** The prompt was reduced strictly to jazz 120 beats per minute. The safety filter passed the input, and audio generation succeeded immediately.

3.2 Identified Potential Variables

1. **V_1 (Controlled Substance String):** The word "acid" in "acid jazz".
2. **V_2 (Geopolitical / Regional Identifier):** The geographic entity "Netherlands" or "Amsterdam".
3. **V_3 (Prompt Density and Context Window):** The cumulative risk score derived from combining multiple descriptive metadata tags (genre + tempo + atmospheric mood + location) in a single request.

4. Hypotheses

Based on the operational failure mode, three testable hypotheses are established:

- **Hypothesis 1 (H_1 - The Narcotics Lexicon Trigger):**
 - *Null Hypothesis ($H_{\{1,0\}}$):* The token "acid" within "acid jazz" is recognized as a musical subgenre and does not trigger safety classifiers.
 - *Alternative Hypothesis ($H_{\{1,A\}}$):* The automated safety system isolates the string "acid" as a prohibited reference to a controlled substance, triggering an immediate execution halt regardless of musical context.
- **Hypothesis 2 (H_2 - The Geopolitical Identifier Trigger):**
 - *Null Hypothesis ($H_{\{2,0\}}$):* Geographic names such as "Netherlands" or "Amsterdam" are benign metadata that do not influence policy compliance scores.
 - *Alternative Hypothesis ($H_{\{2,A\}}$):* Geographic identifiers trigger elevated policy filtering or regional compliance evaluations that result in prompt rejection when combined with atmospheric descriptors.
- **Hypothesis 3 (H_3 - Compound Risk Thresholding):**
 - *Null Hypothesis ($H_{\{3,0\}}$):* Neither V_1 nor V_2 triggers a failure independently; prompt failure occurs strictly due to an additive classifier score across multiple descriptive tokens.
 - *Alternative Hypothesis ($H_{\{3,A\}}$):* The moderation system evaluates prompts using

an additive risk score where V_1 ("acid") and V_2 ("Netherlands") individually pass below the rejection threshold, but combined exceed the safety cutoff.

5. Experimental Testing Strategy

To isolate the failure mechanism, a single-variable controlled experiment must be executed using isolated API prompts.

Test Case	Experimental Prompt String	Target Variable Isolated	Primary Indicator / Evaluation
Test 1	"Acid jazz 120 bpm"	Isolates V_1 (Subgenre Token)	If FAIL : Confirms H_1 ("acid" is an absolute forbidden string). If PASS : Rejects H_1.
Test 2	"Smooth jazz 120 bpm raining in Netherlands"	Isolates V_2 (Geographic Identifier)	If FAIL : Confirms H_2 ("Netherlands" is the trigger). If PASS : Rejects H_2.
Test 3	"Acid house 120 bpm"	Validates V_1 across genres	Tests whether the failure is specific to "acid" across different subgenres.
Test 4	"Groovy funk jazz 120 bpm rainy city mood"	Controlled Baseline (No V_1, No V_2)	Serves as the control group to confirm successful audio synthesis without target tokens.

6. Conclusion and Remediation

The repeated execution failures observed during audio generation highlight a common flaw in multimodal AI pipelines: naive string-matching and context-unaware content moderation. By testing single-variable prompts, system developers and users can pinpoint whether false positives stem from narcotic-related keyword collisions or regional metadata filters. To prevent such failures without sacrificing stylistic intent, generative audio systems must implement domain-aware NLP tokenizers that recognize musical subgenres as atomic entities (e.g., treating "acid_jazz" as a single musical token) rather than parsing sensitive sub-strings independently.

References

1. **arXiv Repository (2026).** *Fine-Tuning Large Language Models for Automatic Detection and Classification of Music Metadata.* arXiv:2602.05485v1.
2. **Google Help Center (2026).** *Generative AI Prohibited Use Policy and Content Guidelines.* Google Support Policies.
3. **iMusician Digital (2026).** *AI False Positives in Music Content Moderation: What Artists Need to Know.* iMusician Industry Insights.
4. **LANDR Audio (2026).** *Prohibited Use Policy for Generative AI Services.* LANDR Legal

Documentation.

5. **ResearchGate Publications (2024).** *Moderating Synthetic Content: The Challenge of Generative AI and False Positive Detection in Audio Workflows.* ResearchGate Legal & Technical Reviews.